

Physics Synthesis and Research Priorities for ASTRA SPPT and SCM

What can be combined and what must remain separate

Source review and bounded mathematical deductions through 5 October 2026

The most defensible consolidation is one common discipline for observation, inference and evidence, supporting distinct physical models. ASTRA can supply that discipline. SPPT remains a planetary transport and inventory program. The Spaghetti Cloud Model (SCM) supplies a geometric evolution to investigate on its own mathematical terms. Operational quantum gravity supplies demanding tests of composition, noise, entanglement and recovery. None of these connections establishes a common microscopic law.

This report brings the supplied research families into a single argument, corrects an important gamma ray source identification, gives compact derivations that sharpen several existing claims, and ranks experiments and mathematical tests by their ability to eliminate ambiguity. It is a research companion, not a promotion of a stable framework or a claim of a new theory of physics.

Evidence convention. Established results, source reported findings, exact conditional deductions, inherited numerical reports and physical conjectures are identified separately. The mathematical checks here are independent algebraic reviews within this AI assisted synthesis; they are not external peer review or experimental replication.

Reading route. The executive findings and research decisions give the main result. Sections on SCM and quantum gravity contain the technical argument. The coverage appendix records a disposition for every topic in the supplied 32 dossier integration, including topics that should remain outside the physics core.

Executive findings

The main opportunity is to reduce conceptual duplication while increasing the specificity of each claim. A common vocabulary is useful only if it prevents a statement about one physical object from silently becoming a statement about another.

- Unify the observation contract and evidence records. A state space, generator, controls, conversion process, instrument, nuisance model, retained record and failure criterion can be specified for every strand. Keep each domain's units, conservation laws and likelihood separate.
- Retain the exact SCM graph mathematics. A compact determinant identity reveals precisely which complex gradient structures produce finite scale corrections. Its small slope Schrödinger limit is a limit of a chosen model, not a derivation of electron matter or universal quantum mechanics.
- Replace broad stability claims with a mode by mode WKI statement. The periodic helical background has an exactly calculable exponentially growing sideband band. Discrete domains may exclude that band, while marginal modes can still have secular growth. A new consistency relation within this analysis links carrier dispersion and sideband growth without the encoding scale.
- Treat nonlinear quantum interpretation as an unresolved completion problem. Norm conservation and Hamiltonian evolution are insufficient for ordinary unitary quantum dynamics. Mixture independence, projective normalization, spectator independence and measurement compatibility must be tested together.

- Separate quantum reconstruction from physical quantum recovery. An informationally complete classical record can identify a state distribution across repeated preparations while losing entanglement with an external reference. This is a decisive boundary between classical identifiability and quantum information recovery.
- Use gravity mediated entanglement as a test of an operational contract. General phase and unequal dephasing formulas are useful, but are already present in recent literature. An entanglement witness violation constrains a declared mediator model and its assumptions; it does not identify a complete ultraviolet theory of gravity.
- Correct the gamma ray input. The supplied news item traces to the H.E.S.S. inner Galaxy line search, not the Fermi nearby subhalo paper used as a topical substitute. Preserve the latter only as a separate comparative calculation. Halo contrast after background fitting, rather than a positive density squared integral alone, controls identifiability in an ON and OFF search.
- Prioritize calibrated inference over expansion into more topics. The strongest empirical next steps are an Enceladus transfer and selection test, a matched evaporation comparison, and an explicit halo nuisance analysis. The remaining literature primarily supplies method benchmarks and counterexamples.

The stable science authority remains SPPT and ASTRA v1.0.7. Later source corrections and research companions should be attached as versioned, unpromoted work. No evidence in this synthesis justifies silently replacing that authority, asserting experimental validation, or publishing an advancement claim.

1 Scope and standards of evidence

The source families reviewed

The review covers the five named October 4 archive families and their scientific manuscripts or ledgers: the research integration supplement, framework update, SCM integration working thesis and supplement, three probe extension and protocols, and quantum gravity dossier. It also incorporates the SCM foundational manuscript, the Convergence package and the WKI carrier correction material. This is coverage of the named artifacts and their accessible scientific content. Membership of every file in a broader project has not been independently established.

The earlier integration supplement contains 19 supplied report dossiers and 13 additional research dossiers. All 32 are assigned a disposition in the coverage appendix. Reviewing that corpus does not mean that every external full text, supplemental file, dataset or calculation has been freshly recovered. Access limitations travel with the scientific claim. In particular, published abstracts, author preprints and final versions are not interchangeable when their wording differs.

Scientific sections and source records were used; personal material is excluded. Existing package test counts, benchmark numbers and replay descriptions are inherited reports unless expressly identified otherwise. No incoming research script was executed for this review, and there was no new experimental data analysis. The new checks concern mathematical entailment, source identity and consistency of the proposed synthesis.

Evidence level	Meaning in this report	What it does not establish
Established primary result	A theorem or published result used within its stated hypotheses; a recent source may instead be identified as a preprint.	Applicability to a new apparatus, ontology or parameter regime.

Evidence level	Meaning in this report	What it does not establish
Source reported observation	What a primary study reports, with access and version limitations.	Independent replication or uniqueness of its interpretation.
Exact within model deduction	Algebra or a proof conditional on explicitly stated definitions and assumptions.	That the model is physically correct or literature novel.
Prior numerical report	A result described in a supplied package or earlier companion.	A fresh execution, robustness check or experimental reanalysis.
Proposed physical interpretation	A hypothesis, adapter, experiment or extension with unresolved obligations.	Admission to the stable science core.

Different kinds of improvement

Three tasks recur and should never share one success flag. A source repair fixes which paper or version supports a sentence. A mathematical repair fixes what follows from a model. An empirical test asks whether a calibrated model predicts observations better than declared alternatives. A clean archive or a passing algebraic identity cannot substitute for the last task. Likewise, an observational anomaly can be real while its favored mechanism remains unidentifiable.

Novelty is treated conservatively. Several compact consequences below are newly integrated or rederived in this review, but a complete literature priority search has not established that they are new theorems. In particular, the arbitrary phase and unequal dephasing entanglement spectrum is explicitly attributed to a September 2026 preprint. The useful advance here is a clearer research program and sharper conditional tests.

2 A common architecture with distinct physical models

The minimal observation contract

Use one contract schema $M = (X, G, U, C, K, N, R, V)$. X is the state space and its physical meaning. G is the generator, domain and boundary conditions. U contains admissible interventions. C converts or propagates the source to the state presented to a detector. K is the observation law and, when relevant, the state update. N contains nuisance parameters and their admissible support. R defines retained records and history. V states what would reject the proposed claim. This is a common schema, not a universal equation of motion.

$$P_u(dy | r, \nu) = \int K_u(dy | z, \nu) C_u(dz | x, \nu) r(dx). \quad (1)$$

For (1), r is a normalized classical source law, and C and K are probability kernels. A daughter configuration may itself be z , allowing fragmentation without losing normalization. If the objective is an expected number of daughters rather than the probability of a full configuration, the object is an intensity measure and must be labeled as such. For quantum data, replace this notation by a complete quantum instrument:

$$P_u(y | \rho) = \text{Tr } I_{y,u}(\rho), \quad \sum_y I_{y,u} \text{ is trace preserving.} \quad (2)$$

An instrument specifies both outcome probabilities and conditional postmeasurement states. The no click, veto and discarded outcomes belong to the complete law whenever the experiment's attempt

denominator is operationally defined. A detected catalogue alone does not create records for events that were never counted. Exposure, trigger counters, censoring and saturation must be preserved where available.

When two models are distinguishable

Let F map a physical description θ to the complete family of outcome laws for the allowed controls. Define θ and θ' to be observationally equivalent when $F(\theta) = F(\theta')$. A target $q(\theta)$ is identifiable precisely when it is constant on each equivalence class. The forward implication is immediate: if q is recovered from F , equal F implies equal q . Conversely, if q is constant on every fiber of F , define $\tilde{q}$ on the image of F by $\tilde{q}(F(\theta)) = q(\theta)$; the value is well defined. This exact set theoretic statement says nothing about finite data precision, conditioning or computational feasibility.

This criterion streamlines several discussions. Different ocean histories can yield the same detected grain distribution. Different halo and particle parameters can yield the same photon prediction. Distinct mixtures can yield the same present density operator. The relevant question is which declared target remains constant over those ambiguities, and which additional intervention splits them. Adding a latent variable helps only when it changes a prediction or resolves a conservation account.

Complementary marginal measurements are not automatically a joint record. Separate instruments can constrain different projections without revealing correlations between them. A purported combined inference must specify whether outcomes concern the same event, the same prepared ensemble, or a matched but distinct population. It must also account for shared calibration and source selection before multiplying likelihoods.

Quantum recovery is a stricter question

An informationally complete positive operator valued measurement (POVM) makes the map from a density operator to its ideal probability law injective. Yet the outcome only channel $M(\rho) = \sum_r \text{Tr}(M_r \rho) |r\rangle \langle r|$ is entanglement breaking. Every decoder that receives only this classical register remains a measure and prepare channel. It cannot generally return an unknown state while preserving its entanglement with a reference. Reconstructing a density matrix from many preparations and physically recovering one input are different operations. [Q8]

For a d dimensional maximally entangled reference input, any separable output has squared fidelity at most $1/d$ with the original maximally entangled state. A replacement channel produces the smaller value $1/d^2$ when the reference is maximally mixed. Thus a channel may have perfectly informative classical statistics and still fail a reference preserving recovery requirement. This distinction prevents classical observation fibers from being misused as a solution to quantum information loss.

What should actually be shared

Shared component	Domain specific component	Admission check
Claim identity and version	Physical variables and units	Statement and source version remain traceable.
Forward model interface	Constitutive laws and generators	Every interface has a declared domain and codomain.
Nuisance and selection bookkeeping	Detector response and likelihood	Calibration support covers the proposed inference.

Shared component	Domain specific component	Admission check
Alternative and rejection records	Mechanism specific countermodels	A named alternative makes a measurably different prediction.
Validation status	Experiment and recovery criterion	Reported, derived, simulated and observed remain separate.

SPPT should retain its reservoirs, phase transformations, transport, stoichiometry and energy inventories. SCM should retain its graph state, geometric Hamiltonian and analytic evolution. Quantum gravity contracts should retain mediator locality, channel structure, conserved quantities and reference systems. Similar diagrams cannot identify a plume kernel with a quantum generator or turn a state geometry into an electron material model.

3 The geometric SCM model and its interpretation

The exact graph equation

Let $U = (u, v)$ be a smooth map from a Euclidean base M to an oriented two dimensional plane, and let $F(x) = (x, U(x))$. The base may be periodic with a fixed volume measure. The induced graph metric is $g_{ij} = \delta_{ij} + \partial_i U \cdot \partial_j U$. While a smooth graph chart exists, skew mean curvature flow with the appropriate tangential reparametrization gives a Hamiltonian evolution in graph coordinates. The tangential correction matters: the vertical graph velocity is not the uncorrected ambient binormal velocity. [B3, G1, G2]

$$i \partial_t \psi = -\beta \partial_i (\sqrt{\det g} g^{ij} \partial_j \psi), \quad \psi = u + iv. \quad (3)$$

With $\beta = \hbar/(2m)$, small graph slopes make $\sqrt{\det g} g^{-1}$ approach the identity, recovering the free scalar Schrödinger equation. Matching β to $\hbar/(2m)$ is a parameter identification. It does not derive the electron mass, charge, spin, Pauli exclusion, electromagnetic coupling or many particle composition. Smoothness, boundary conditions and the lifetime of the graph chart remain independent analytic obligations. General global well posedness in the physically relevant dimension cannot be inferred from formal conservation laws.

A compact invariant controls the finite scale correction

Write the graph encoding as $U = a(\text{Re } \psi, \text{Im } \psi)$, with a a declared dimensional scale. If ψ is unit normalized in d spatial dimensions, a^2 has dimensions length to the power $d+2$; writing $a^2 = \ell^{d+2}$ makes the slope $a \nabla \psi$ dimensionless. Use the fixed Euclidean base metric to define $T = \nabla \psi^* \cdot \nabla \psi$ and the complex quadratic contraction $\Xi = \nabla \psi \cdot \nabla \psi$. The star denotes complex conjugation. Both are local functions; $|\Xi|$ is not the density $|\psi|^2$. This is a scalar pure graph construction, not a definition on arbitrary density operators.

$$\det g = 1 + a^2 T + (a^4/4)(T^2 - |\Xi|^2) \quad (4)$$

$$= (1 + a^2 T/2)^2 - a^4 |\Xi|^2/4. \quad (5)$$

Derivation. Let $r = \nabla \text{Re } \psi$ and $s = \nabla \text{Im } \psi$. The only nonzero perturbations of the identity metric lie in their span. Sylvester's identity reduces $\det g$ to the determinant of a two by two Gram matrix: $1 + a^2(|r|^2 + |s|^2) + a^4(|r|^2|s|^2 - (r \cdot s)^2)$. Since $T = |r|^2 + |s|^2$ and $\Xi = |r|^2 - |s|^2 + 2i r \cdot s$, equation (4) follows. The identity is exact in any base dimension for this codimension two scalar encoding.

The area Hamiltonian, with the flat reference contribution removed, can be normalized as follows:

$$H_a[\psi] = (\hbar^2/ma^2) \int (\sqrt{\det g} - 1) dx. \quad (6)$$

$$H_0[\psi] = (\hbar^2/2m) \int T dx. \quad (7)$$

Rationalizing the difference gives an exact nonnegative kinetic deficit:

$$H_0 - H_a = (\hbar^2 a^2/4m) \int |\Xi|^2 / [1 + a^2 T/2 + \sqrt{\det g}] dx \geq 0. \quad (8)$$

The exact energy H_a is nevertheless nonnegative, since g is the identity plus a positive semidefinite matrix and $\sqrt{\det g} \geq 1$. In the controlled small slope regime $a^2 T \ll 1$, its expansion is

$$H_a = H_0 - (\hbar^2 a^2/8m) \int |\Xi|^2 dx + O(a^4). \quad (9)$$

The order symbol assumes adequate control of the higher gradient terms under integration; it is not uniform over arbitrary high frequency normalized states. The quartic truncation in (9) is unbounded below when extrapolated to arbitrarily large gradients. That extrapolation is outside the approximation and does not prove the exact graph Hamiltonian is unbounded below.

An exact flux identity avoids matrix inversion

$$\sqrt{\det g} g^{-1} \nabla \psi = [(1 + a^2 T/2) \nabla \psi - (a^2 \Xi/2) \nabla \psi^*] / \sqrt{\det g}. \quad (10)$$

Equation (10) follows by multiplying its numerator by g and obtaining $(\det g) \nabla \psi$. It is algebraically useful because the nonlinear flux needs only two scalar contractions and no d by d matrix inverse. It is regular at a node $\psi = 0$ when the gradients are smooth; divisions by the wavefunction have not been introduced. Expanding the flux and restoring $\hbar$ yields

$$i\hbar \partial_t \psi = -(\hbar^2/2m) \Delta \psi + (\hbar^2 a^2/4m) \nabla \cdot (\Xi \nabla \psi^*) + O(a^4). \quad (11)$$

This is a practical streamlining of both analysis and a future implementation, not an executed implementation change. If Ξ vanishes identically on a whole spatial slice, the exact flux equals $\nabla \psi$ and the energy equals H_0 . Balanced rank two gradients therefore hide this finite scale correction. Sensitivity depends on the chosen state family. However, $\Xi = 0$ is not generally a dynamically preserved constraint, so equality at an instant is not a theorem that every such initial state follows the free equation for all time. Equality at one point alone does not make the differentiated generators equal there.

Why this invariant does not define a mixed state theory

The quantity Ξ is built from a chosen pure wavefunction. A density operator can have inequivalent ensemble decompositions, so averaging a nonlinear pure state energy or flow branch by branch needs an additional rule. A useful counterexample uses three orthonormal equal energy plane waves with pairwise orthogonal wavevectors in a periodic box. Their equal mixture has the same density operator as the equal mixture of a discrete Fourier rotated orthonormal basis. For the original basis the summed $|\Xi|^2$ integral is three times the corresponding sum for the rotated basis. Thus the quartic correction averaged over components depends on the decomposition despite the same density operator. [B3; exact conditional check]

This does not prove every possible nonlinear theory inconsistent. It proves that the pure graph invariant cannot simply be relabeled a decomposition independent density operator functional. A physically meaningful completion must state whether the ensemble decomposition is operationally real, whether ordinary density matrices remain complete statistical states, and how classical records and remote preparations act on the dynamics.

A separate proposed completion constructs $C_{ij}[\rho] = \text{Re } \partial_{xi} \partial_{vj} \rho(x,y)|_{y=x}$ directly from a density kernel. Unlike the pure scalar tensor, this can have rank greater than two. For the equal mixture of three orthogonal axial plane waves in $d = 3$ on volume V , $C[\rho] = \lambda I_3$ with $\lambda = k^2/(3V)$. Then the proposed area kinetic density is $(\hbar^2/ma^2)[(1+a^2\lambda)^{3/2}-1]$, which is strictly greater than its linear counterpart $3\hbar^2\lambda/(2m)$ for $\lambda>0$. This reverses the pure state inequality. It is an exact counterexample to importing (8) into that mixed state completion, not a contradiction of the pure graph result.

The quantum consistency gates

Gate	What must be checked	Consequence for the present program
Transition probabilities	A common map must preserve overlaps if ordinary pure state unitarity is claimed.	Norm conservation alone is insufficient.
Mixtures and records	Equivalent density operators and forgetting classical flags must give compatible predictions.	Branchwise nonlinear evolution needs a new measurement and composition rule.
Projective normalization	Multiplying a representative by an irrelevant scalar must not change a claimed ray law.	A fixed physical a and a normalization convention cannot be left implicit.
Independent spectators	Product states must not change an active subsystem through arbitrary spectator normalization volume.	The naive many particle graph extension is not automatically acceptable.
Boost covariance	The stated laboratory symmetry must survive the finite scale terms or be explicitly broken.	The standard Galilean boost gate fails in the supplied finite scale model.
Well posedness	Specify function space, boundary domain, graph lifetime and norm/energy balance.	A formal geometric expression is not a global evolution theorem.

For a concrete mixture obstruction, take $f_0=1/\sqrt{L}$ and $f_k=\exp(ikx)/\sqrt{L}$, with $\Omega_0=\beta k^2$ and $\eta=a^2 k^2/L$. The exact constant plus carrier family evolves with mode frequency $\Omega(r)=\Omega_0/\sqrt{1+\eta r}$, where r is the population of f_k . Define $\chi_1=\sqrt{p} f_0+\sqrt{(1-p)}f_k$ and $\chi_2=\sqrt{(1-p)}f_0-\sqrt{p} f_k$. The equal mixtures of $\{f_0,f_k\}$ and $\{\chi_1,\chi_2\}$ have the same initial density operator. Branchwise evolution changes the latter because its two carrier populations produce different frequencies. This family is invariant under the full continuum PDE; it is not a numerical mode truncation. [B3]

The density difference is $[[0,c],[c^*,0]]$, hence its trace distance is $D=|c|=\sqrt{p(1-p)}|\sin(\Delta\Omega t/2)|$, with $\Delta\Omega=\Omega(1-p)-\Omega(p)$. In the simultaneous $\eta\ll 1$ and $|\Delta\Omega t|\ll 1$ regime, $D\approx\Omega_0\eta|t|\sqrt{p(1-p)}|1-2p|/4$. It is maximized at $p=(1\pm 1/\sqrt{2})/2$, giving $D\approx\Omega_0\eta|t|/16$. This is an optimized measurement distinguishability estimate conditional on branchwise evolution and ordinary density operator measurement rules, not a measured signaling effect.

Three interpretations should remain explicit. As an encoding, the graph can illuminate ordinary state geometry. As a nonlinear classical field, it can be studied without claiming a quantum ontology. As a fundamental modification of quantum mechanics, it must meet the gates above or provide a coherent replacement. A material electron interpretation adds further burdens and is not established by any of these interpretations alone.

The minimal record test makes the assumption explicit. For a flagged preparation $\sum_i p_i |i\rangle\langle i| \otimes \rho_i$, suppose each branch obeys the same isolated evolution, the record remains classical, the reduced density operator is a sufficient state, and forgetting the record commutes with local evolution. These assumptions require affinity:

$$\Phi_t(\sum_i p_i \rho_i) = \sum_i p_i \Phi_t(\rho_i). \quad (12)$$

A nonlinear density commutator can preserve positivity, trace and eigenvalues of an individual state while failing this condition. A stronger no signaling argument additionally invokes the allowed bipartite preparations, local operations and ordinary trace probability rules. Such conditional arguments do not exclude every altered nonlinear operational theory, or stochastic nonlinear trajectories whose averaged density evolution remains affine. [G4, G5, G6]

Time reversal and the choice of Hamiltonian

For the autonomous scalar graph equation with real coefficients and a compatible closed boundary domain, complex conjugation together with $t \rightarrow -t$ gives scalar time reversal covariance. Its square is +1. This does not supply spin one half time reversal, which acts on a different state space, and it does not imply spontaneous time reversal breaking. A complex field or a ninety degree normal rotation is not itself a time reversal odd order parameter. [B4]

Likewise, a codimension two wavefunction graph and a codimension two nodal submanifold are different geometric objects. Curvature, torsion, normal rotation and filament flow depend on the actual Hamiltonian and parameterization. Hasimoto's relation between vortex filament motion and a nonlinear Schrödinger equation is an important exact connection under its own construction; it does not license assigning that nonlinear equation to every geometric representation of a wavefunction. [G2]

The Poisson structure is part of that construction. Near a straight graph the Hasimoto field scales as a second spatial derivative of the transverse displacement. Pulling back the canonical wave symplectic form therefore introduces derivatives into the symplectic form. Keeping the ordinary graph form while substituting the curvature based energy changes the principal operator: an energy proportional to the squared third derivative generates a sixth order rather than second order flow. A correct equivalence must transform the bracket as well as the Hamiltonian and handle zero modes, rigid motions and closure. [G8]

There is also a known higher dimensional obstruction to a naive Hasimoto extension. Khesin and Yang show that the Willmore quantity $\int |H|^2$ need not be conserved under higher dimensional skew mean curvature flow; their Clifford torus example demonstrates the failure explicitly. They also show that nonzero normal curvature obstructs a globally integrated torsion phase. Thus a curvature amplitude construction cannot inherit the one dimensional normalized NLS interpretation without further structure. This does not invalidate the separate graph height PDE. [G9]

4 Carrier dispersion and sideband growth in the WKI reduction

The model and the perturbation convention

In one base dimension with zero potential, the graph equation becomes a potential form of the Wadati Konno Ichikawa equation. Differentiation with $w = a\partial_x\psi$ gives the established complex WKI equation. The equations below use laboratory coordinate x and laboratory time t with fixed $\beta > 0$. They are not automatically the same perturbation problem as one posed after a reciprocal or moving coordinate transformation. [G3, B6]

$$i\psi_t = -\beta \partial_x[\psi_x / \sqrt{(1+a^2|\psi_x|^2)}], \quad (13)$$

$$iw_t = -\beta \partial_x^2[w / \sqrt{(1+|w|^2)}]. \quad (14)$$

On a periodic interval of length L , w has zero spatial mean, and integration recovers ψ only up to a constant. The potential equation conserves that constant mode when the potential vanishes. Consider $\psi_0 = A \exp[i(kx - \omega t)]$, with $A > 0$, $k = 2\pi m/L \neq 0$, slope intensity $s = a^2 A^2 k^2$ and $g = 1+s$. Its exact frequency is

$$\omega = \beta k^2 / \sqrt{g}. \quad (15)$$

Let $\delta\psi = \exp[i(kx - \omega t)](u + iv)$, with small real u and v . The radial and tangential derivatives of the nonlinear flux differ. Keeping both gives the first order system

$$u_t + c u_x = -\beta v_{xx} / g^{3/2}, \quad (16)$$

$$v_t + c v_x = \beta u_{xx} / \sqrt{g} + \beta k^2 s u / g^{3/2}, \quad (17)$$

$$c = \beta k(g+1)/g^{3/2}. \quad (18)$$

For a perturbation $\exp(\lambda t + iQx)$, eliminating u and v produces the exact linear dispersion relation

$$(\lambda + icQ)^2 = \beta^2 Q^2 [k^2 s - gQ^2] / g^3. \quad (19)$$

Thus an exponentially growing sideband exists precisely when $0 < Q^2 < k^2 s / (1+s)$. The real growth rate is the positive square root of the right side of (19). This is an exact property of the linearization about the specified carrier, not a nonlinear stability theorem and not an experimental result.

Finite periodic domains alter the test

$$Q = 2\pi n/L, \quad 0 < n^2 < m^2 s / (1+s). \quad (20)$$

For $|m| = 1$ there is no exponentially growing nonzero periodic sideband at finite s . For $|m| \geq 2$ the first permitted sideband enters the exponentially growing band exactly when $s > 1/(m^2 - 1)$. At equality the nonzero cutoff has a nontrivial Jordan block, permitting linear in time growth. The $Q = 0$ amplitude direction also gives secular phase drift; fixed norm removes that direction for a pure carrier, but a constant plus carrier state can compensate its norm through the constant component. The phrase no exponentially growing sideband is therefore deliberately narrower than stable.

Differentiation maps the coupled carrier sideband pair through factors $ia(k+Q)$ and $ia(k-Q)$. Both are nonzero throughout the exponentially unstable band because $|Q| < |k|$. Potential and zero mean slope descriptions therefore have conjugate linearized operators on that band under the same smooth periodic boundary conditions. The exceptional $Q = \pm k$ sectors are outside the band and contain the constant potential information lost by differentiation. This settles a limited potential versus slope question, not the entire arclength, frame and closure problem.

A scale eliminated consistency relation

Define $r = \omega/(\beta k^2) = 1/\sqrt{1+s}$, with β independently calibrated. Algebraically eliminating s from (18) and (19) gives a joint prediction:

$$\gamma(Q)^2 = \beta^2 Q^2 r^4 [k^2(1-r^2) - Q^2], \quad (21)$$

Equation (21) gives squared real exponential growth only inside the unstable band. Outside it the real growth rate is zero; the negative radicand describes oscillatory eigenvalues, not a negative squared real growth rate.

$$c = \beta k r(1+r^2), \quad (22)$$

$$\gamma_{\max} \text{ continuum} = (\beta k^2/2)r^2(1-r^2) \leq \beta k^2/8. \quad (23)$$

Once phase suppression is fixed, the model predicts sideband cutoff, growth and drift without fitting a second encoding scale. On a periodic lattice the physical maximum is over allowed integers n , and generally falls below the continuum maximum. The global upper bound in (23) is attained at $s = 1$ only if $|m|$ is even and $|n| = |m|/2$. A unit normalized pure carrier also obeys $A^2 L = 1$, hence $s = a^2 k^2/L$; its amplitude is not an independent free control at fixed a , L and k .

A lone carrier frequency is a global phase, so an experiment must specify a relative phase reference before calling r observable. The exact constant plus carrier family $B + A \exp[i(kx - \omega t)]$ supplies one mathematical reference, since B leaves ψ_x unchanged and remains constant in zero potential. Its amplitude fraction and normalization must be stated, and any constant potential must be calibrated or subtracted. In an analogue field test, measure phase response and then predict held out seeded sideband growth and drift, including ordinary dispersion, interactions, phase noise and selection in the baseline. Passing this test would support this PDE in that apparatus; it would not establish a fundamental electron theory.

The literature comparison is still open

The 2019 WKI source prints the relevant equation, but also repeats a no modulation instability statement associated with modified WKI literature. The older transformed background and perturbation conventions were not fully recovered in this review. The direct periodic calculation should be preserved with its explicit coordinates and boundary conditions. It should not be advertised as resolving a contradiction in the literature or discovering an entirely new instability. A proper comparison must carry the background, spectral parameter, periodicity and time map through the transformation. [G3, G7]

5 Operational quantum gravity and information

Entangling phase is a contract level observable

A gravity mediated entanglement (GME) proposal begins with assumptions about two accessible subsystems, their initial separability, permitted interactions, mediator properties and complete readout. A local classical mediation channel implemented by local operations and classical communication (LOCC) cannot create entanglement from a separable input without another quantum resource. That is a statement about the operational channel. Calling a potential classical because it has numerical coefficients does not by itself establish that a full two system channel obeys those premises.

The Newtonian branch phase, a nonclassical mediator conclusion and evidence for propagating gravitons are separate claim levels. They require different exclusions of nongravitational interactions, matter exchange, cross talk and selection. A low energy gravitational effective field theory provides a controlled

language for specified amplitudes and scales; a visual similarity between an SCM graph metric and spacetime is not such a derivation. [Q4, Q11]

General phase and unequal dephasing

Assume initially balanced factorized path qubits, a deterministic diagonal branch unitary, and independent local Z dephasing with real coherence factors a and b in $[0,1]$. Here a and b are coherence factors and are unrelated to the graph encoding scale in Section 3. Removing local phases leaves

$$\rho = U\delta [(I+aX)/2 \otimes (I+bX)/2] U\delta^\dagger, \quad (24)$$

$$U\delta = \exp(i\delta Z \otimes Z/4), \quad \delta = \varphi_{00} + \varphi_{11} - \varphi_{01} - \varphi_{10}. \quad (25)$$

The four eigenvalues of the partial transpose are

$$\lambda(\varepsilon, \pm) = [1 + \varepsilon ab \pm \sqrt{(a^2+b^2+2\varepsilon ab \cos\delta)}/4], \quad \varepsilon = \pm 1. \quad (26)$$

This general spectrum is already given in the September 2026 preprint by Rizaldy and colleagues, with an equivalent dephasing notation. It is rederived and integrated here, not claimed as new. For two qubits the positive partial transpose criterion is necessary and sufficient for separability. Only the $\varepsilon = -1$ lower branch can become negative, although it is not always the smallest eigenvalue in the separable region. [Q3]

$$N = \max\{0, [\sqrt{((a-b)^2 + 4ab \sin^2(\delta/2))} - (1-ab)]/4\}, \quad (27)$$

$$\text{Entanglement} \Leftrightarrow 4ab \sin^2(\delta/2) > (1-a^2)(1-b^2). \quad (28)$$

A short proof also fixes the domain of the result. Let $c = \cos(\delta/2)$ and $s = \sin(\delta/2)$. Pauli conjugation gives $4\rho = I + a(cXI - sYZ) + b(cIX - sZY) + abXX$. Partial transpose on the second qubit flips the ZY term. All terms commute with XX. In its ε sector, $IX = \varepsilon XI$ and $ZY = \varepsilon YZ$, while XI and YZ anticommute. The resulting two by two spectra give (26). The $\varepsilon = +1$ lower branch is nonnegative because its square root is at most $a+b$, and $a+b \leq 1+ab$.

At $\delta = \pi$, (28) reduces to $(1+a)(1+b) > 2$. At $a=b=1$, $N = |\sin(\delta/2)|/2$. At $\delta=0$ or $a=0$ the state is separable. These controls should be compulsory checks for any implementation or quoted threshold. A special phase threshold should not be applied when the accumulated phase is unknown or fluctuating.

A sharp constant rate consequence

Now make the additional assumptions $a(T) = \exp(-\gamma_a T)$, $b(T) = \exp(-\gamma_b T)$, with constant nonnegative rates, and $\delta(T) = \kappa T$. Dividing (28) by $4ab$ gives the exact condition

$$\sin^2(\kappa T/2) > \sinh(\gamma_a T) \sinh(\gamma_b T). \quad (29)$$

$$\text{There exists } T > 0 \text{ with entanglement} \Leftrightarrow |\kappa| > 2\sqrt{\gamma_a \gamma_b}. \quad (30)$$

Necessity follows from $\sin^2(\kappa T/2) \leq \kappa^2 T^2/4$ and $\sinh(\gamma_a T) \sinh(\gamma_b T) \geq \gamma_a \gamma_b T^2$. If the strict rate inequality holds, the leading T^2 terms imply (29) for sufficiently small positive T , establishing sufficiency. Equality never suffices. If one rate vanishes, some entangling time exists exactly when κ is nonzero. For equal positive rates γ , any entangling time must also satisfy $T < \text{asinh}(1)/\gamma$.

Within this narrow model, waiting longer cannot rescue a rate below the threshold. Above the threshold, the result does not guarantee a detectable signal, feasible preparation time, a successful particular witness or a gravitational origin. Correlated dephasing, phase jitter, amplitude damping, unbalanced preparation, changing geometry and non Markovian rates require a different calculation. The scientific value of (30) is a bounded design veto, not a universal noise limit.

Keep every outcome and its denominator

For a quantum instrument, the unnormalized flagged map $\Phi(\rho) = \sum_r |r\rangle\langle r| \otimes I_r(\rho)$ is affine and trace preserving when all outcomes are included. The normalized conditional state $I_r(\rho)/\text{Tr } I_r(\rho)$ is generally nonlinear. This familiar conditional nonlinearity does not license a fundamental nonlinear evolution: the full record process still obeys ordinary affinity. [Q2]

The same bookkeeping matters for metrology. Let success have parameter dependent probability $p(\theta)$, a conditional success density $q(y|\theta)$, and an uninformative failure flag. Under regularity conditions, information per attempted preparation is

$$F_{\text{full}} = [p'(\theta)]^2/[p(1-p)] + p F_{\text{success}}. \quad (31)$$

If failures also have informative records, add $(1-p)F_{\text{failure}}$. The identity follows by factoring the joint likelihood into flag probability and conditional outcome, whose score has zero mean. A loss factor multiplying conditional information is valid only when survival is parameter independent and failures carry no additional information. The quantum analogue decomposes information into the classical flag contribution and weighted conditional quantum Fisher information. Preparation, successful detection and wall clock cost may favor different protocols; the denominator must be stated. [Q2]

Changing measure does not derive gravity

For a prescribed positive measure $\mu(t,x)dx$ and a Hamiltonian self adjoint in $L^2(\mu dx)$, a convenient norm preserving equation is

$$i\hbar[\partial_t + \frac{1}{2}\partial_t \ln \mu]\psi = H\psi. \quad (32)$$

The half density transform $\chi = \sqrt{(\mu/\mu_0)}\psi$ removes the explicit measure connection relative to a fixed reference measure, while transforming H . It does not remove physical time dependence in the Hamiltonian. More generally, a positive metric operator $G(t)$ on the state space admits a connection A satisfying $A^\dagger G + GA = \dot{G}$. Norm preservation fixes only that symmetric part; one may add a G anti Hermitian term. A chosen symmetrized action can select a connection, but normalization alone does not derive the dynamics or a gravitational field equation. [B5]

Representations must transform together. For $\chi = \sqrt{\mu}\psi$, the derivative $D_i\chi = (\partial_i - \frac{1}{2}\partial_i \ln \mu)\chi$ restores $\mu^{-1}\text{Re}[(D_i\chi)^* D_j\chi] = \text{Re}[(\partial_i\psi)^* \partial_j\psi]$. Transform the state, generator, subsystem algebra and instrument before checking whether the record law changed. Transforming only the state while freezing the detector can manufacture an apparent physical effect. A Euclidean graph metric, a probability measure and a dynamical Lorentzian metric remain distinct objects.

Clock records need a multitime instrument

Relational clock constructions can yield conditional unitary states from a stationary correlated history under specified constraint, spectrum and inner product assumptions. Uniform clock counts alone do not establish the necessary correlations. Nor does a family of one time states specify a multitime measurement process. Identity evolution and an intervening Z pulse both preserve an initial $I/2$ at sampled endpoints, but two X measurements give equal outcomes in the first experiment and opposite outcomes in the second. [Q1, Q5]

The accessible apparatus must also be explicit. Measuring X and Z separately does not distinguish $\rho_\pm = (I \pm rY)/2$, for $0 < r \leq 1$, even though X and Z generate the full matrix algebra under formal multiplication. Algebraic closure does not grant a Y measurement or a joint instrument. Use the actual effect or instrument span for observational equivalence unless operational closure is justified. A clock contract

therefore needs intervention order, measurement updates, clock error, clock system interaction, memory, and conditioning probabilities.

Recovery needs a target and an error norm

The reset channel shows why output purity is insufficient: a pure replacement can erase all dependence on the input. Section 2 gives the stronger distinction between statistical tomography and reference preserving recovery. For a proposed recovery R after channel E , a valid entangled benchmark is $\varepsilon\Phi = \frac{1}{2} \|[(R \circ E) \otimes \text{id}](\Phi d) - \Phi d\|_1$, with Φd maximally entangled. It is different from the worst case error $\varepsilon\Diamond = \frac{1}{2} \|R \circ E - \text{id}\|_\Diamond$.

For the fixed tested composite $R \circ E$ given by $U = \text{diag}(-1, 1, \dots, 1)$, the squared entanglement fidelity is $((d-2)/d)^2$ and $\varepsilon\Phi = 2\sqrt{(d-1)/d}$, which tends to zero as d grows. Yet $(|0\rangle + |1\rangle)/\sqrt{2}$ is mapped to an orthogonal state, so $\varepsilon\Diamond = 1$. A good normalized Choi benchmark therefore supplies no dimension independent worst case guarantee for that fixed recovery. This is not irrecoverability: choosing U^{-1} repairs the channel. At fixed finite dimension, exactly zero Choi error still implies channel equality.

Recovery existence, available coherent access and efficient implementation are three distinct questions. Sufficiency and information disturbance theorems constrain specified states or codes, but do not automatically give a physically accessible efficient decoder for astrophysical radiation. A classical transcript cannot be treated as a coherently available quantum radiation subsystem. A report should state the code, observable algebra, reference, norm, copy resources, decoder access and complexity assumptions before invoking a recovery claim. [Q8, Q9, Q10]

The classical gravity debate remains disputed

The published Aziz and Howl proposal is challenged on how interactions, matter exchange and conditioning enter the model. An April 2026 preprint argues that restoring discarded transition amplitudes restores factorization. A July response disputes that calculation's treatment of remaining particle overlap factors and interprets the mechanism as quantum matter exchange with postselection. The April objection must not be presented as an uncontested final refutation. [Q12, Q6, Q7]

The durable experimental lesson is to keep leakage and postselection sectors, establish subsystem independence, and test entangling nongravitational channels. Proposed species and barrier comparisons need matched controls because barriers can also change electromagnetic backgrounds. An entanglement witness constrains the full contract actually implemented; stronger statements about mediator ontology or gravitons require additional assumptions and observables. No controlled experimental detection was established by this bounded source review, and that review is not a proof of universal absence.

6 Empirical adapters that can test the architecture

Gamma rays require a source correction

The exact supplied gamma ray news item can now be linked to the H.E.S.S. Inner Galaxy Survey paper by Aharonian and colleagues, Physical Review Letters 137, 091002, DOI 10.1103/d8tj-55kc, arXiv 2608.07234. The news page is dated 4 October 2026 despite a September segment in its URL. The earlier three probe manuscript transparently used the nearby subhalo Fermi study by Salces Pérez and colleagues as a topical substitute because that mapping was unresolved. The new information changes the target, instrument, energy range and likelihood. The source identity must be superseded explicitly, not silently renamed. [E1, E2]

H.E.S.S. reports 546 hours of Inner Galaxy observations and no significant line signal. Its search uses spatial and energy ON and OFF regions, both containing modeled dark matter signal. Its reported line rate is $\langle\sigma v\rangle_{\text{line}} = \langle\sigma v\rangle_{\gamma\gamma} + \frac{1}{2}\langle\sigma v\rangle_{\gamma Z}$. The primary study's limits are conditional on the adopted Milky Way density profile and response. The source correction is not a dark matter detection and not empirical validation of ASTRA. Event records, masks and run specific response files were not acquired in this review.

The Fermi case remains scientifically useful as a separate comparison. Its previously recorded Navarro Frenk White (NFW) integral $J = 1.15115074 \times 10^{23} \text{ GeV}^2 \text{ cm}^{-5}$ is an assumed profile numerical calculation, not a replay performed here and not a reproduction of the H.E.S.S. analysis. Its stated source distance and complete halo radius also do not meet a generic far target bound requiring $R < D$. Selection geometry, observer centered line of sight convention and confidence construction must be checked before reusing its numerical limit. [E2]

A positive density squared integral does not guarantee sensitivity

For nonnegative density ρ over a declared region, Cauchy Schwarz gives $\int \rho^2 dV \geq M^2/V$ when captured mass M and volume V are fixed. More generally, for positive weight w with finite $\int w^{-1} dV$, the weighted bound is $\int w \rho^2 dV \geq M^2 / (\int w^{-1} dV)$. Concentrating a fixed mass can make the square integral arbitrarily large unless additional density or regularity restrictions are imposed. These are geometric inequalities, not a unique astrophysical halo reconstruction. [B4]

A rate upper bound derived from a signal proportional to θJ needs a calibrated lower support bound on the relevant J , as well as an observation model. If nuisance support allows J to approach zero, a finite robust upper bound on θ does not follow from a product limit alone. Conversely, a large positive J can still be observationally invisible when the signal is aligned with a fitted background. The latter point is especially relevant to the corrected ON and OFF problem.

Consider an idealized single bin with independent Poisson counts and positive interior means. Let $\alpha > 0$ be a known OFF to ON background ratio, b an unknown background intensity, θ a rate, and s_{ON} and s_{OFF} fixed response folded unit rate signal templates:

$$\mu_{\text{ON}} = b + \theta s_{\text{ON}}, \quad \mu_{\text{OFF}} = \alpha b + \theta s_{\text{OFF}}. \quad (33)$$

$$I\theta \text{ after profiling } b = (\alpha s_{\text{ON}} - s_{\text{OFF}})^2 / (\mu_{\text{OFF}} + \alpha^2 \mu_{\text{ON}}). \quad (34)$$

To derive (34), use the two parameter Poisson information matrix. Its determinant is $(\alpha s_{\text{ON}} - s_{\text{OFF}})^2 / (\mu_{\text{ON}} \mu_{\text{OFF}})$, while $I_{bb} = 1/\mu_{\text{ON}} + \alpha^2/\mu_{\text{OFF}}$. Their quotient is the Schur complement for θ . If $s_{\text{OFF}} = \alpha s_{\text{ON}}$, then $b' = b + \theta s_{\text{ON}}$ absorbs the rate exactly and local information vanishes. Sensitivity is controlled by response projected contrast after nuisance fitting, not by positive J alone.

Equation (34) is a newly checked analytic illustration, not the actual multi bin H.E.S.S. likelihood or a diagnosis of failed H.E.S.S. sensitivity. Its α convention is the reciprocal of some ON and OFF conventions. Unknown α , correlated systematics, boundaries, finite count coverage and shape varying halo nuisances require the full model. A scalar J interval is not enough when a nuisance changes morphology or ON and OFF contrast.

One authoritative Enceladus adapter

The framework update and the three probe extension describe the same source to record problem and should be merged. The state is an ocean or reservoir composition plus vent history and a parent droplet distribution. Conversion includes phase partition, chemical reaction, fragmentation, transport and loss. The observation law includes impact speed, size, composition, triggering, telemetry, storage and

classification. The target might be an ocean elemental abundance or a mass flux. It must not be replaced by a detected grain category frequency. [B2, B4]

Postberg and colleagues report chemically diverse salt rich grains and propose a slow freezing and fragmentation reconstruction. The primary abstract and institutional account support the source identity and bounded interpretation. The complete final article and supplement were not independently retrieved here. The institutional release's 961 spectra and illustrative laboratory example are not an event membership manifest or a universal natural threshold. The rapid freezing conference work remains a preliminary motivation for an alternative pathway, not an established refutation. [E3, E4]

For fragmentation, use a probability law on the complete daughter configuration, or a branching random measure. Its first moment $F(dz|x)$ integrates to expected multiplicity and can predict intensities; it does not encode sibling correlations or replace the full likelihood. Conservation should be written for elements and all relevant reservoirs, allowing molecular reactions. Grain number, detected counts and record files do not obey the elemental mass balance.

If each item has known positive inclusion probability π_i and a measured mass or elemental mark m_i , inverse inclusion weighting can give an unbiased total $\sum \text{selected } m_i/\pi_i$ under its sampling assumptions. A normalized ratio of two such estimated totals is not automatically unbiased. Zero response classes remain unidentified. If mass and detection efficiency are inferred from the same impact waveform, their errors are correlated and need a joint measurement model rather than independent plug in weights.

The NASA PDS Cassini CDA archive is a verified data route, but it does not identify the exact 961 spectrum subset or reconstruct the incident grain denominator. A newly located velocity calibration study uses IID QI signals, not the CAT mass spectra used for Type 3 chemistry. That study is a useful separate calibration lead; importing its predictor as the missing chemistry response would cross an unverified instrument boundary. [E5, E6]

The best experiment crosses a known source recipe with droplet size and pressure or cooling history, randomizes preparation batches, images material before and after freeze drying, and records parent daughter mass plus same parent composition covariance. Calibrate an actual impact instrument and any laser analogue separately. Hold out entire preparation histories and predict joint daughter marks. A fitted pooled histogram is too weak if different histories produce the same histogram.

Evaporation is a scoped comparison

The positive photomolecular interpretation and the later flat water null should share one contradiction record with separate apparatus domains. The null reports no measurable enhancement under continuous visible illumination near solar irradiance for the tested wavelengths and flat neat water configuration, with an approximately 4.8 percent reported rate sensitivity. That value is an apparatus detection threshold, not a universal equivalence margin or a proof against every light assisted evaporation mechanism. [E7, E8]

A matched test needs common geometry, absorbed power, temperature fields, humidity, air flow, polarization, wavelength, mass metrology and independent session replicates. It should predict both evaporation and thermal records. Freeze a scientifically meaningful effect or equivalence margin before testing, and propagate correlated metrology rather than treating every time sample as an independent replicate. Failure to reject a difference is not positive evidence of equivalence. The deposited data route for the null paper is a promising first readiness check; its data were not recovered or reanalyzed here.

Superconductivity does not complete the scalar electron model

The published YbSb₂ abstract reports spontaneous field evidence and proposes an internally antisymmetric nonunitary triplet interpretation with possible Majorana surface modes. The final abstract should govern the outward claim strength; the accessible January author version and final paywalled article are separate records. Pairing nonunitarity concerns a pairing matrix and is not failure of unitary time evolution. A scalar complex phase and the graph normal rotation are not a microscopic explanation of this material. [E9]

A sole nondegenerate leading instability of a real symmetric quadratic Ginzburg Landau form can be chosen real. Intrinsic relative complex order at a continuous onset therefore needs a demonstrated degeneracy, additional transition or other mechanism under the material's actual symmetries. The relevant calculation should use orthorhombic spin, orbital and spin orbit representations, rather than promoting an illustrative cubic Dirac model to a material specific topological proof.

A second analysis of the same muon curves is not an independent experimental data root. The accessible author version also reports a longitudinal decoupling field around 10 mT, greater than its quoted type I critical field near 51 G, or 5.1 mT. This motivates a phase boundary aware control: if the applied field suppresses superconductivity, a relaxation change alone cannot establish the field dynamics of an otherwise unchanged superconducting state. This is an inference caveat about the accessible version, not a refutation of the zero field onset evidence.

Protected Majorana claims require a specified Bogoliubov de Gennes symmetry class, invariant and boundary protection mechanism. Enumerate symmetry allowed mass terms for the actual candidate and surface termination. If an allowed perturbation gaps a crossing, that protection claim fails while other observations may remain valid. Generic fully gapped classification does not rule out every crystalline, nodal or weakly protected alternative. A complementary odd or phase sensitive measurement must have a calculated coupling before a null result can reject the order. [E10]

Shared nuisance means a shared witness

There is a small but important correction to combining experiments. Suppose noiseless $y_1 = \theta + v$ and $y_2 = \theta - v$. Each alone leaves θ unrestricted if v is unrestricted, but together they identify $\theta = (y_1 + y_2)/2$ if the same v applies. Profiling a separate nuisance for each experiment discards that information. Conversely, pretending that a drifting calibration is shared creates spurious information.

The joint identified set is $\{\theta: \text{there exists one common } v \text{ compatible with all experiment laws}\}$, with additional domain specific nuisances where justified. It is not generally the intersection of individually projected sets that may use incompatible nuisance witnesses. Exact observational equivalence is well defined on complete states; mere overlap between nuisance supported outcome sets need not be an equivalence relation. This distinction belongs in the common inference core because it affects photon response, muon calibration and laboratory grain recipes alike.

7 A smaller and more discriminating research program

The consolidation decisions

Current duplication or conflation	Recommended disposition	Reason
Enceladus framework adapter and three probe Protocol C	Merge into one branching and elemental mass adapter.	The later protocol supplies physics missing from simple count scaffolds.
Repeated product aliases and rank tests	Use one identified target chapter with typed examples.	They illustrate the same inference limitation without sharing a physical law.
Several geometric encodings	Retain a map ledger with domain, boundary and Poisson structure.	Beam, graph and Hasimoto maps have different objects and dynamics.
Pure graph and density kernel completion	Keep distinct model specifications.	Rank and energy inequalities differ; records remain an open gate.
GME, clock and recovery front matter	Use one operational contract cover with three separate protocols.	Preparation, full records and errors are reusable; conclusions are not.
Gamma source and NFW integral	Replace source mapping; retain Fermi calculation as comparison.	The requested H.E.S.S. instrument is a different experiment.
Source status and numerical pass labels	Separate source report, exact proof, code test and replication.	No single established or passed label can carry all meanings.
Biology, archaeology and organizational analogies	Retain a short method appendix.	They do not add physical support for an ASTRA or SCM mechanism.

Ranked next tests and stopping rules

Priority and target	Evidence needed	Completion or demotion rule
0 Repair the gamma contract	Correct H.E.S.S. source, line definition, ON and OFF response, explicit Fermi comparison.	Complete when each numerical statement points to the correct experiment and version.
1 Freeze the SCM operational interpretation	One state and mixture law, normalization convention, composite rule and full instrument.	Either satisfy the flag and spectator tests or label the model classical or effective with altered assumptions.
2 Test the carrier consistency relation	An independently calibrated β , actual boundary modes, phase reference and held out sideband predictions.	Reject the chosen PDE if one parameter choice cannot jointly explain phase, drift and growth within the declared error model.
3 Establish Enceladus data readiness	Exact event subset, trigger history and appropriate CAT calibration support.	Proceed only with a verified readiness inventory; otherwise report the missing identifying information.

Priority and target	Evidence needed	Completion or demotion rule
4 Perform material specific symmetry analysis	Actual YbSb ₂ representations, allowed GL terms and boundary masses.	Retain only onset and protection claims supported without undeclared tuning or forbidden perturbations.
5 Reconstruct the evaporation null	Deposited data, uncertainty model and matched apparatus controls.	Report the scoped bound or identified set; freeze a justified margin before a new equivalence experiment.
6 Stress the GME contract	Full outcomes, local and correlated noise, exchange and cross talk countermodels.	A witness supports only the mediator conclusions surviving those checks.
7 Compare recovery criteria	Reset, measure and prepare, invertible unitary and phase flip controls across code size.	Never replace a stated reference or worst case target by purity or an average score.

These priorities combine tractability with information gained. The source repair can be completed without new data. The SCM consistency gates can close an overbroad interpretation mathematically. The remaining empirical tasks require accessible data or new experiments and should not be reported complete after a synthetic check. A calibrated null or a rigorously bounded identified set is a successful scientific outcome when it answers the declared question.

What would justify promotion

Promotion should require a bounded claim with units and scope; a fixed source version; a specified forward law; known exact aliases; calibrated nuisance support; a comparison with a simpler baseline; an independent or held out observation; and an explicit failure action. A proposed new variable should earn its place by changing an observable prediction, closing an inventory or distinguishing alternatives. A positive simulation with the same assumptions used to generate its target is a consistency check, not an independent evidence root.

The first meaningful empirical milestone should be one correctly reproduced domain result with calibrated uncertainty and an honest failure region. Only then is it worth expanding adapters. More symbolic parallels, module names or synthetic perfect classifications cannot substitute for that milestone. Stable v1.0.7 authority should remain intact until a separately authorized scientific release decision is made.

8 A clear account for a wider physics audience

The most general lesson is that physics does not observe an underlying state without a chain of interactions. A grain from an ocean can freeze, fragment and be selected before its spectrum is recorded. A dark matter distribution can produce photons that must be separated from foregrounds. A quantum state can leave a classical record while losing the entanglement needed to recover it. Understanding the chain is part of understanding the phenomenon.

Geometry can also connect equations without making their objects identical. A curve, a graph of a complex field and a quantum state can share elegant mathematics, while differing in what is measured, what evolves and what is conserved. The graph model reviewed here has exact mathematical content and a familiar small slope limit. Its finite scale terms produce further predictions and also expose conflicts

with ordinary mixture and composition rules. Those conflicts are useful because they tell us exactly which interpretation needs repair.

The strongest progress is therefore narrower and more testable than a claim of unification. One can simplify the geometry to a few invariants, derive a joint dispersion and growth test, correct which gamma ray experiment is being discussed, and specify what an entanglement or recovery experiment would actually establish. None of this solves quantum gravity, identifies dark matter, derives an electron, proves a biosphere on Enceladus or demonstrates a new force. It improves the questions so that the next calculation or observation can genuinely change the answer.

This public research draft is unpromoted and has not received external specialist or peer review. Its mathematical deductions are conditional on the stated models. Stronger novelty or physical validation claims require further source, specialist and empirical review.

Appendix A Coverage and disposition of the supplied research

The following directory makes the coverage boundary explicit. The named scientific artifact families were reviewed for their claims, connections and conflicts. The 32 topic dispositions below are an integration audit of the supplied source records. They are not 32 new full text reviews or independent replications. Topic IDs are retained so that future corrections can be made without changing which study a statement originally meant.

Source family	Scientific role	Disposition
October research integration	32 topic intake and eight proposed bridge contracts.	Retain as the breadth register; reduce the active empirical portfolio.
October framework update	Observation and claim structure; Enceladus and evaporation protocols.	Consolidate repeated inference objects; keep stable authority separate.
SCM working thesis and supplement	Exact graph equations and quantum consistency gates.	Retain mathematics and conditional obstructions; simplify via invariants.
Three probe extension and protocols	Gamma rays, superconductivity and Enceladus.	Correct gamma source, merge physical adapters and preserve failure criteria.
Quantum gravity dossier	GME, relational clocks and recovery contracts.	Update literature and sharpen reference, noise and full record tests.
SCM foundational manuscript	Geometric encoding and proposed physical interpretation.	Use as a model specification with unresolved interpretation gates.
Convergence package	Geometric and operational synthesis, including alternative maps.	Retain exact maps with their brackets and domains; remove physical overidentifications.
WKI carrier correction	Reference and boundary condition question.	Integrate the direct periodic result and leave literature reconciliation open.

The nineteen supplied reports

H01 Enceladus transformed samples combine and prioritize

Merge with three-probe Protocol C into one branching/calibration flagship. Paper-specific event membership and CAT chemical response are the immediate gate.

Source locator from supplied register: <https://www.science.org/doi/10.1126/sciadv.aee7256>

H02 Inka coca deposits retain only as an archive example

Trace presence, deposition and ritual reconstruction are distinct. No new ASTRA physical module is justified; defer full interpretation pending methods and preservation context.

Source locator from supplied register: <https://doi.org/10.1080/00776297.2026.2719320>

H03 focused light fluctuations retain as a controlled distribution benchmark

Full enhancement distributions can distinguish correlations missed by a mean. Keep finite-size, channel control and drift nulls; do not combine its forward model with H09 photonic topology.

Source locator from supplied register: <https://www.nature.com/articles/s41467-026-77933-y>

H04 flat water evaporation null combine with D06

One contradiction/support audit and one registered equivalence protocol. Reproduce stated observational precision before designing a new cross-laboratory experiment.

Source locator from supplied register: <https://www.pnas.org/doi/10.1073/pnas.2615377123>

H05 Antarctic conditional loss retain as a model discrepancy example

Emulator calibration and common simulation ancestry are the reusable obligations. Defer a new adapter until withheld observations and omitted-physics alternatives are specified.

Source locator from supplied register: <https://www.nature.com/articles/s41561-026-02102-1>

H06 group attachment and dissent move to governance appendix

Exploratory hypothetical-task evidence does not demonstrate scientific correction quality or independent AI review. Do not count it as physical framework support.

Source locator from supplied register: <https://www.nature.com/articles/s41598-026-65147-7.pdf>

H07 70 Oph orbital constraints retain as a claim level calibration example

Allowed stable orbit, detectable planet and observed planet are distinct predicates. Actual inclination/activity injection-recovery is needed for an application.

Source locator from supplied register: <https://arxiv.org/html/2603.20044v2>

H08 local pH electrosynthesis retain as an optional reaction transport case

Local proton activity cannot be replaced by bulk pH. Defer quantitative scale-up claims until final methods, balances and causal response controls are accessible.

Source locator from supplied register: <https://pubs.acs.org/jacsat/article/doi/10.1021/jacs.6c12994/5421141/Local-pH-Control-Promotes-Oxygen-Atom-Transfer>

H09 lossy photonic readout retain as a separate instrument encoding adapter

Keep normalization, dilation, coherence zeros and reconstruction conditioning explicit. Do not promote emulator observables into material topology or quantum resources.

Source locator from supplied register: <https://journals.aps.org/prl/accepted/10.1103/xkht-64kc>

H10 atmospheric momentum and rotation combine with D11

One covariance-aware angular-momentum/inertia budget with common epochs and external torque. Scenario outputs and present attribution must remain distinct.

Source locator from supplied register: <https://www.nature.com/articles/s41612-026-01382-z.pdf>

H11 tokay calls retain as a proxy validation example

Acoustic/pose evidence and inferred laryngeal mechanism are separate. No direct connection to SCM dynamics; defer mechanistic strengthening pending direct functional measurement.

Source locator from supplied register: <https://pubmed.ncbi.nlm.nih.gov/42772768/>

H12 Galactic centre X ray mixtures retain as an inverse mixture stress test

Challenge component count, weak-line stability and alternate foregrounds. “Uncontaminated” must not be imported as a theorem of unique decomposition.

Source locator from supplied register: <https://arxiv.org/html/2604.07588v1>

H13 slow fast funnels combine methodologically with D07

A full/reduced basin-validity benchmark is useful. It needs boundary refinement and finite-timescale tests; do not conflate classical phase-space tunneling with quantum tunneling.

Source locator from supplied register: <https://journals.aps.org/prl/abstract/10.1103/jtkh-9lz5>

H14 YbSb2 combine with three probe Protocol B

Retain the field/pairing/topology ladder and phase-aware LF control. Do not use this material to validate a fundamental scalar SCM electron law.

Source locator from supplied register: <https://journals.aps.org/prl/abstract/10.1103/drzq-lfn5>

H15 MEGATRON simulated histories retain as a forward to observation case

Related progenitors and snapshots are dependent; simulated spectral diversity is not observed unique ancestry. Pair with D05 only through an explicit mock-observation layer.

Source locator from supplied register: <https://arxiv.org/html/2510.05201v2>

H16 DarkSide ultraheavy null retain as a distinct exclusion domain example

Composite scattering, overburden and detector selection define a different likelihood from LZ and gamma annihilation. No pooled “dark-sector exclusion” object.

Source locator from supplied register: <https://arxiv.org/html/2602.09421v1>

H17 squid lateral line anatomy combine only in a sensory evidence ladder

Anatomy, function and cross-species translation are separate stages. No therapeutic inference or new physical module follows.

Source locator from supplied register: <https://pubmed.ncbi.nlm.nih.gov/42767195/>

H18 double neutron star timing retain as a theory conditioned parameter example

Masses and decay consistency depend on the timing theory and kinematic corrections. It is not a direct observation of frame dragging or a general QG discrimination.

Source locator from supplied register: <https://arxiv.org/html/2607.27333v2>

H19 disk hosted black hole growth retain as a population selection example

Bulgeless morphology is an imperfect history proxy. Defer causal growth pathways until selection, resolution and independent historical alternatives are modeled.

Source locator from supplied register: <https://arxiv.org/html/2609.22434v1>

The thirteen additional studies

D01 Bennu soluble chemistry retain direct reservoir sample comparison

Useful for contamination, inventory and processing checks. It supplies no Enceladus transfer kernel, biosignature or evidence of biological origin.

Source locator from supplied register: <https://www.nature.com/articles/s41550-024-02472-9>

D02 DESI DR2 dark energy retain separately as conditional model comparison

Dataset combinations, priors and cosmological parameterization define the preference. No shared likelihood with dark-matter particle searches and no new universal field implied.

Source locator from supplied register: <https://arxiv.org/html/2503.14738v3>

D03 LZ WIMP limits retain a version qualified null benchmark

Keep the v3 number with v3 assumptions and power constraint. Combine only reporting discipline with H16; their particle response supports differ.

Source locator from supplied register: <https://arxiv.org/html/2410.17036v3>

D04 GW230529 retain object identity uncertainty example

Posterior component masses and object classification are distinct. A preferred black-hole interpretation is not definite identity from the GW data alone.

Source locator from supplied register: <https://arxiv.org/html/2404.04248v3>

D05 early galaxy oxygen line combine with H15 s mock observation question

A detected emission line supports prior enrichment, while total elemental mass and population history require an excitation/selection model.

Source locator from supplied register: <https://arxiv.org/html/2409.20549v2>

D06 photomolecular interpretation combine with H04 retaining both records

A positive proxy and a scoped null need matched support before contradiction. Mechanism specificity requires molecular/thermal controls beyond flux alone.

Source locator from supplied register: <https://www.pnas.org/doi/10.1073/pnas.2320844121>

D07 adaptive network canards combine with H13 s reduction benchmark

Stability of an overall cycle and its component saddle states are different. Retain the adaptation law and finite regime, not a cosmological analogy.

Source locator from supplied register: <https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.133.237401>

D08 non Abelian photonic design retain as an encoding caution

Proposed classical pseudospin braiding and physical non-Abelian anyons/topological qubits are not interchangeable. Defer a new adapter until an actual device/readout benchmark is available.

Source locator from supplied register: <https://journals.aps.org/prl/abstract/10.1103/rgfy-n6zd>

D09 non Hermitian geometry combine at the instrument audit level with H09

Gauge/basis choice, tensor definition and response regime must remain explicit. Do not merge with SCM metric geometry on verbal similarity.

Source locator from supplied register: <https://journals.aps.org/prresearch/abstract/10.1103/qb8s-9c6y>

D10 gecko vibration pathway retain stronger functional comparator for H17

Physiology in one species can inform experiment design but cannot calibrate another species' sensor response or human therapy.

Source locator from supplied register: <https://www.sciencedirect.com/science/article/pii/S0960982224012272>

D11 climate mass redistribution combine with H10

One physical angular-momentum budget, with scenario and reconstruction periods, common inputs and units preserved; no addition of headline length-of-day rates.

Source locator from supplied register: <https://pubmed.ncbi.nlm.nih.gov/39008671/>

D12 active hair bundle thermodynamics retain optional energetic boundary case

An active feedback model needs its local free-energy supply and omitted dissipation explicit. Whole-organ energy efficiency is not the modeled subsystem efficiency.

Source locator from supplied register: <https://journals.aps.org/prxlife/abstract/10.1103/6wcm-z333>

D13 ancient peach management retain only an archive inference comparator with H02

Use, management, cultivation and domestication are distinct historical propositions. Preserve ecological and taphonomic alternatives rather than creating another ASTRA mechanism.

Source locator from supplied register: <https://profiles.wustl.edu/en/publications/evidence-for-a-prolonged-pre-domestication-phase-8000-year-old-pe/>

The resulting portfolio has two near term empirical flagships, Enceladus and matched evaporation; one corrected gamma benchmark; one material specific superconductivity gate; several queued domain tests; and a concise set of inference counterexamples. This preserves the useful lessons while avoiding the appearance that every new headline adds a new physical mechanism.

Appendix B Source access and references

References B identify the supplied scientific manuscripts, not external validation. References G, Q and E identify primary literature or official scientific data resources used for the focused synthesis. Access notes are intentionally explicit where a published full text, dataset or version was not newly examined. The bibliography supplies direct locators, while the body states the supported claim and its limitations. Source retrieval and algebraic verification do not imply a fresh code replay.

Supplied scientific manuscripts

[B1] ASTRA SPPT Research Integration, integration supplement v0.1.0, 4 October 2026. Scientific manuscript and 32 source dossiers. Basis for the full topic coverage directory.

[B2] ASTRA SPPT Framework Update v0.2.0 and Source Verification companion, 4 October 2026. Observation architecture, proposed contracts and prospective Enceladus and evaporation protocols.

[B3] Quantum State Geometry and Observation Contracts in ASTRA and SPPT, working thesis v0.2.0, and research supplement, 4 October 2026. Scientific manuscript and mathematical gates.

[B4] Hidden States and Discriminating Observations, thesis extension v0.3.0, and Protocols v0.3.0, 4 October 2026. Three probe mathematical and empirical contracts.

[B5] ASTRA Quantum Gravity Dossier QG 0.1.0, 4 October 2026. Changing measure, gravity mediated entanglement, clock and recovery constructions.

[B6] Potential WKI Sidebands A Reference and Boundary Condition Question, v0.1. Two page correction note. The direct periodic calculation was independently checked in this synthesis.

[B7] ASTRA Convergence Master, 4 September 2026, especially the geometric synthesis. Used with the separate map and Poisson qualifications in this report.

[B8] SCM Quantum State Geometry foundational preprint v0.1.0. Supplied mathematical model and proposed interpretation; not an independently validated electron theory.

Geometry and quantum foundations

[G1] Song and Sun. Skew mean curvature flow. Regime specific existence result for compact surfaces in $\mathbb{R}^4$; not a global all data theorem in three base dimensions. <https://arxiv.org/abs/1502.04525>

[G2] Hasimoto. A soliton on a vortex filament. Journal of Fluid Mechanics 51 (1972), 477–485. Original curvature and torsion construction. <https://www.cambridge.org/core/journals/journal-of-fluid-mechanics/article/soliton-on-a-vortex-filament/1A2FF58DD1B8DE20509866A4713531F3>

[G3] Zhang and colleagues. Physica D (2019), DOI 10.1016/j.physd.2019.05.008. Accessible publisher equation and introduction checked; older stability convention remains unresolved. <https://www.sciencedirect.com/science/article/abs/pii/S0167278918303907>

[G4] Jordan. Classical properties imply quantum dynamics. The classical correlated record assumptions behind affinity are essential. <https://arxiv.org/abs/quant-ph/0508092>

[G5] Simon, Bužek and Gisin. No Signaling Condition and Quantum Dynamics. Conditional derivation using standard quantum statistical assumptions. <https://arxiv.org/abs/quant-ph/0102125>

[G6] Polchinski. Weinberg's nonlinear quantum mechanics and the Einstein Podolsky Rosen paradox. Physical Review Letters 66, 397 (1991). <https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.66.397>

[G7] Modified WKI study, Theoretical and Mathematical Physics (2017), DOI 10.1134/S0040577917050117. Primary record located; transformed background conventions not fully recovered. <https://www.mathnet.ru/eng/tmf9197>

[G8] Holm and Stechmann. Hasimoto transformation and vortex filament hierarchy, especially Section 5 on the Poisson map and reparametrization. <https://arxiv.org/abs/nlin/0409040>

[G9] Khesin and Yang. Nonlinearity 34, 1525–1542 (2021), DOI 10.1088/1361-6544/abe1d1. Proposition 3.6, Example 3.7 and Corollary 4.4 checked for higher dimensional curvature and torsion obstructions. <https://www.math.utoronto.ca/khesin/papers/hasimotoNonlinearity.pdf>

[G10] Khesin and Modin. 2026 preprint on Madelung geometry with noncritical zeros. A mathematical lead with explicit topology and regularity hypotheses. <https://arxiv.org/html/2607.01024v1>

Operational quantum information and gravity

[Q1] Pollock and colleagues. Operational Markov condition for quantum processes (2018). Multitime instrument and process distinction. <https://arxiv.org/abs/1801.09811>

- [Q2] Combes, Ferrie, Jiang and Caves. Quantum limits on post selected probabilistic quantum metrology. *Physical Review A* 89, 052117 (2014). Flag information and full resource denominator.
<https://arxiv.org/abs/1309.6620>
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- [Q4] Donoghue. Quantum General Relativity and Effective Field Theory. arXiv 2211.09902v2 (2023). Retained from the supplied dossier; not a new full paper audit in this pass.
<https://arxiv.org/html/2211.09902v2>
- [Q5] Höhn, Smith and Lock. The Trinity of Relational Quantum Dynamics, arXiv 1912.00033; and Equivalence of approaches to relational quantum dynamics in relativistic settings, *Frontiers in Physics* 9, 587083 (2021). Spectral and sector assumptions matter.
<https://www.frontiersin.org/journals/physics/articles/10.3389/fphy.2021.587083/full>
- [Q6] Gundhi, Infantino and Bassi. Can classical theories of gravity produce entanglement? arXiv 2604.19696v1, 21 April 2026. Preprint objection, contested by Q7.
<https://arxiv.org/html/2604.19696v1>
- [Q7] Tibau Vidal and Iyer. Matter Field Exchange Generates Entanglement Not Classical Gravity. arXiv 2607.03429v1, 3 July 2026. Exchange, postselection and remaining overlap critique.
<https://arxiv.org/html/2607.03429v1>
- [Q8] Horodecki, Shor and Ruskai. General Entanglement Breaking Channels (2003). Theorem 4 gives the measure and prepare characterization used here. <https://arxiv.org/abs/quant-ph/0302031>
- [Q9] Junge, Renner, Sutter, Wilde and Winter. Universal recovery maps and approximate sufficiency of quantum relative entropy. *Annales Henri Poincaré* 19, 2955 (2018). <https://arxiv.org/abs/1509.07127>
- [Q10] Kretschmann, Schlingemann and Werner. The Information Disturbance Tradeoff and the Continuity of Stinespring's Representation. Scope checked; no numerical theorem constants imported.
<https://arxiv.org/abs/quant-ph/0605009>
- [Q11] Mitakos, Papageorgiou, Perche and Christodoulou. When does entanglement through gravity imply gravitons? arXiv 2601.03214v2, pinned revision 15 April 2026.
<https://arxiv.org/html/2601.03214v2>
- [Q12] Aziz and Howl. Classical theories of gravity produce entanglement. *Nature* 646, 813–817 (2025); subsequent preprint version arXiv 2510.19714v3. The claim under dispute, not a settled inference.
<https://www.nature.com/articles/s41586-025-09595-7>

Empirical adapter sources

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- [E2] Salces Pérez and colleagues. Nearby subhalo Fermi LAT line search. arXiv 2609.03016v2. Retained solely as a comparative source, not the primary behind the supplied news.
<https://arxiv.org/html/2609.03016v2>
- [E3] Postberg and colleagues. Cassini CDA observes compositional segregation of Enceladus ice grains from slow freezing and fragmentation of oceanic spray. *Science Advances* 12, eade7256 (2026). Final full article and supplement access remains incomplete. <https://doi.org/10.1126/sciadv.aee7256>

[E4] Hogan and colleagues. Segregation of Enceladus ocean salts within rapidly frozen droplets at vent relevant pressures. EPSC 2026 abstract 1326. Preliminary conference evidence; detailed experimental statements inherited from the supplied manuscript.

<https://meetingorganizer.copernicus.org/EPSC2026/EPSC2026-1326.html>

[E5] NASA Planetary Data System. Cassini CDA bundle version 1.0, DOI 10.26033/6esc-cn13. Archive route verified; exact paper event subset not recovered.

https://pds.nasa.gov/ds-view/pds/viewBundle.jsp?identifier=urn:nasa:pds:cassini_cda&version=1.0

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<https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2026EA005130>

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10.1073/pnas.2615377123. Primary full text reviewed; deposited data DOI 10.17617/3.Q2GHPJ not recovered. <https://www.pnas.org/doi/10.1073/pnas.2615377123>

[E8] Lv, Tu, Zhang and colleagues. Photomolecular effect Visible light interaction with air water interface. PNAS 121, e2320844121 (2024), DOI 10.1073/pnas.2320844121; author preprint arXiv 2310.19832v1. Positive interpretation and apparatus scope retained separately.

<https://www.pnas.org/doi/10.1073/pnas.2320844121>

[E9] Kataria, Srivastava, Samanta and colleagues. Observation of Time Reversal Symmetry Breaking in the Type I Superconductor YbSb₂. Physical Review Letters 137, 136002 (2026), DOI 10.1103/drzq-lfn5. Published abstract and author version arXiv 2601.07460v1; final full text and supplement access incomplete. <https://journals.aps.org/prl/abstract/10.1103/drzq-lfn5>

[E10] Schnyder, Ryu, Furusaki and Ludwig. Classification of topological insulators and superconductors in three spatial dimensions. Primary classification and its symmetry assumptions.

<https://arxiv.org/abs/0803.2786>

Additional official source linkage: the H.E.S.S. collaboration overview supports the corrected news mapping at <https://hess-experiment.eu/2026/08/HESS-SOM32028/>. The Enceladus institutional release is

https://www.elsi.jp/en/news_events/highlights/2026/enceladus_ocean_spray_diverse_ice_grains/.

Releases are used for source linkage and explicitly attributed examples; they do not replace primary methods.

Appendix C Planetary synthesis and final research priorities

Research edition dated 5 October 2026

Saturn and the Earth Moon system strengthen the case for a common discipline of calibrated inference. They do not supply a common microscopic dynamics for SPPT and SCM. The most useful new connections concern what survives transport, which phase of a response is measured, when a present inventory ceases to be a clock, and how an instrument selects a population. Each leads to a bounded test that can fail.

This addendum completes the planetary literature synthesis while retaining the preceding mathematical argument and its qualifications. Established mathematics, source reported observations, model dependent interpretations and proposed tests remain separate. Publication dates below distinguish recent analysis from the often much older spacecraft measurements. The literature review includes independent source and algebra checks; it includes no fresh execution of archived research code, event level reanalysis or experimental replication.

C1 What the new evidence changes

- SPPT gains a sharper separation of elemental inventory, phase conversion, material transport and observational selection. Shell thickness, optical cleanliness and a detected grain or ion fraction are different targets.
- ASTRA gains concrete benchmarks for joint response measurements and full exposure records. Titan couples response amplitude and lag; lunar negative ions expose conversion versus detector calibration; Saturn plasma surveys show why event counts need coverage.
- SCM retains the exact conditional graph and WKI results in Sections 3 and 4. Planetary waves and filaments are comparison cases for testing a proposed generator. No source reviewed here maps a planetary observable to SCM's state or measures its predicted carrier sideband relation.

The proposed connections are newly assembled research designs in this synthesis. A literature priority claim is not established. Their scientific value is the extra observable, constrained nuisance or rejection criterion they supply, rather than a resemblance between shapes or equations.

C2 Saturn ring cleanliness and the loss of an age clock

Source reported model. Ricerchi and Crida's July 2026 Icarus study revises the exposure age calculation by treating planar gravitational focusing, retained impactor material and persistence of optical absorption separately. Their model permits a pollution equilibrium that loses age information. It does not establish that Saturn's rings are ancient or solve their origin and dynamical survival. The 2023 Cassini based young exposure age inference remains a competing, assumption dependent result. [P1, P2]

Source reported observation with a modeled origin. Linti and colleagues reported an off-plane silicate dust population in November 2025, compositionally compatible with ring material. Ring ejection and electromagnetic transport are interpretations requiring size and detection assumptions. This is a possible independent constraint on material loss, not a direct measurement of the efficiency that controls an optical exposure clock. Final paper and supplement access was incomplete. [P3]

Exact illustrative deduction. Let x be a dilute contaminant mass fraction in a ring of constant total mass, initially $x=0$. Let $a>0$ be constant retained contaminant input per ring mass per unit time, and $\lambda\geq 0$ a constant physical removal rate. Both a and λ have units of inverse time. In the domain where the mass fraction approximation is valid, the equation $dx/dt=a-\lambda x$ gives, for $\lambda>0$,

$$x(t) = (a/\lambda)[1 - \exp(-\lambda t)]. \quad (P1)$$

The lossless apparent age x/a is no greater than the true t within this toy model. For known a and λ , the inverse is

$$t = -\ln(1 - \lambda x/a)/\lambda, \quad 0 \leq \lambda x/a < 1. \quad (P2)$$

The limit $\lambda\rightarrow 0$ is $x=at$. As x approaches a/λ , the age becomes ill conditioned: a small inventory error corresponds to a large age range. For an all-time literal mass fraction model, a/λ must not exceed one; otherwise this approximation must stop before it leaves its physical range. These statements follow directly by solving the linear equation and differentiating the inverse. They are conditional mathematics, not a new ring age estimate or a replacement for the published evolution model.

The distinction between physical and optical loss is essential. In this toy model, measuring outgoing contaminant mass flux per ring mass gives $j=\lambda x$ and, for $x>0$, constrains λ . It still cannot recover age without the retained input a . Chemical alteration can instead remove an optical signature without removing the corresponding element. A dust flux cannot calibrate that optical coefficient by itself. Variable ring mass, unknown initial pollution and changing impact flux add further ambiguity.

Proposed test. Jointly predict optical contamination, off-plane elemental spectra and atmosphere-bound material flux under one impact and retention model. Measure optical alteration and physical mass loss separately in laboratory analogues. Hold out latitude bands or observing epochs. Reject a claimed age determination if a broad age range survives the independently justified source, loss and detector uncertainties. CDA impact spectra, exogenic trajectory events and Enceladus salt-rich samples need separate response models and denominators.

C3 Titan and the Saturn decagon demand joint dynamical predictions

Source reported measurement. Petricca and colleagues, published 17 December 2025, reanalyzed Cassini radio tracking and obtained $\text{Re}(k_2)=0.608\pm 0.048$ and $\text{Im}(k_2)=0.135\pm 0.035$. These quoted errors are covariance-derived 1σ uncertainties; their Figure 1 displays 3σ bars. The joint response favors the tested ocean-free, dissipative high-pressure-ice models over the tested global-ocean models. The interpretation permits liquid pockets. It is not a model independent exclusion of all subsurface liquid. [P4]

Proposed connection and test. A large response amplitude can share several interior explanations; the dissipative quadrature supplies another constraint. This is a physical example of splitting observation-equivalent models with a complementary measurement. SPPT should jointly account for phase state, rheology, heat generation and heat export. Compare fixed model families against the covariance-aware complex tidal response, rotation and independent seismic predictions. A new interior parameter earns its place only if it changes one of these predictions. Full replay also requires the declared tracking data and orbit-model infrastructure, including the research licence required by MONTE.

Source reported observation and model. The Saturn decagon paper appeared electronically on 2 September 2026. Its institutional author abstract reports a roughly $58\text{--}63^\circ$ S wave moving eastward at 2.5 m/s alongside a jet peaking at 116 m/s, with vertex-longitude oscillations around 32 days. Shallow-water models offer candidate forcing mechanisms. Hubble detections extend back to 2023, but changing visibility prevents assigning that date as a demonstrated physical onset. Full primary article and supplement access was incomplete. [P5]

Proposed connection and test. Moving wave geometry and moving material are distinct observables. Test a rotating-fluid model against pattern speed relative to the measured jet, vertex oscillation, wavelength and temporal evolution together. This parallels the discipline of SCM's joint carrier-phase, drift and growth test in Section 4 without importing its generator into Saturn's atmosphere. A polygonal image alone supplies neither a dispersion law nor evidence for SCM.

C4 Enceladus and brines separate phase history from recorded composition

Source reported model. Gilkyson and colleagues, published 11 September 2026, study brine entrainment during ice-shell growth. Their conductive NaCl model shows how liquid incorporation can increase geometric shell thickness without corresponding basal freezing or latent-heat release at the ice-ocean interface. Later freezing within the shell can release that heat. It is a tidal-free endmember, omitting radiogenic and tidal heating, convection and later mushy compaction. Its synthetic moon classes and salinity profiles are model baselines, not measured reconstructions of Enceladus, Dione, Mimas or Europa. [P6]

Proposed connection. An inventory model must record solidification, retained liquid, expelled brine and salt precipitation separately. Geometry alone cannot close the latent-heat budget. The same unresolved liquid inclusions can change radar attenuation, so fitting an electromagnetic response does not by itself identify salinity. A useful laboratory test measures solidification, liquid incorporation, elemental totals and transmission on the same sample series, then predicts held-out cooling histories and independently characterized inclusion geometries.

Access update to Section 6. The Postberg slow-freezing and fragmentation paper was published on 25 September 2026; the linked news page displays 4 October. Its abstract reports about 1,000 Type-3 spectra, while the institutional release specifies 961. The final methods, supplement and selected-event manifest remain unavailable. The EPSC rapid-decompression abstract has now been read in full rather than inherited from the earlier manuscript. It reports salt segregation in recovered freeze-dried residues and remains preliminary conference evidence. Those maps do not determine which features arose before drying. [E3, E4]

Complementary laboratory evidence. Hamp and colleagues' September 2026 study distinguishes NaCl-water phase assemblages that depend jointly on salinity and cooling history. Cryogenic Raman spectroscopy and diffraction provide phase information, but the near-infrared experiment found no unique SC2-II bands; later warming can erase a metastable phase. Chinnery and Fox-Powell's October 2025 experiments show different sublimation behavior for equal-composition material frozen under different histories. Their 10 weight percent analogues expose a microstructural effect rather than calibrate natural grains; extension to E-ring sputtering is untested. [P18, P19]

Exact conditional implication. If two equal-composition preparations have different future output laws under the same later exposure, composition alone is not a sufficient predictive state on that support. An averaged composition-only law may fit one mixture of histories and fail when that mixture changes. SPPT should therefore test whether phase assemblage, microstructure or a constrained latent history distribution improves held-out predictions. This is a classical state-sufficiency question; it neither identifies mineral texture with a wavefunction nor resolves SCM's mixture problem.

Proposed discriminating experiment. Cross known recipes and parent sizes with slow cooling, rapid decompression and mixed histories. Measure chemistry and phase before drying, then again after controlled drying or warming. Close the elemental balance over gas, solids, liquid and wall deposits, including carbon loss. Hold out whole recipe-history combinations. Phase and chemistry must concern the same grain or a declared matched population. No Cassini CAT capability to identify these

crystallographic phases has been established here; specify a phase-sensitive laboratory measurement or separately calibrated future instrument.

A standard branching control sharpens the count test. Suppose parent arrivals in a fixed exposure are Poisson with mean $\Lambda > 0$. Independent identically distributed parents make K daughters, with finite second moment and positive mean. Each daughter is independently detected with the same probability $p > 0$; exclude dead time, saturation, source-rate drift and additional correlations. If N is the detected total, then

$$\text{Var}(N)/E[N] = 1 + p E[K(K-1)]/E[K]. \quad (P3)$$

Derivation. Binomial thinning gives detected daughters per parent mean $pE[K]$ and second factorial moment $p^2E[K(K-1)]$. A compound Poisson sum has variance Λ times the per-parent second moment, yielding (P3). Low efficiency suppresses this particular Fano-factor excess, so nearly Poisson detected counts do not rule out physical fragmentation. Conversely, excess variance is not unique evidence of siblings: variable source rate or response can produce it, while detector timing can distort it. This is standard conditional probability, not a new theorem or a Cassini measurement.

Use tagged parent droplets, measured collection efficiency, blanks and timing controls to test first and second moments before adopting a full marked-cluster likelihood. Physical survival and detector selection must remain separate stages. A future fresh-plume versus aged-grain comparison needs a shared source model but separate aging, speed and recording-mode responses. The newly checked IID velocity-calibration article explicitly leaves CAT extension to future work; a lower usable waveform threshold does not recover impacts that were never triggered or telemetered. [E6]

C5 Plasma structures require a coverage aware transport model

Source reported observation. Hadid and colleagues' Enceladus Alfvén-wing study was first published on 9 February 2026. It identifies 127 geometry-selected eligible encounters, 36 with clear MAG signatures, including 13 non-flyby crossings. Reflected structures extend at least about 120° downstream. The ratio 36/127 is conditional on the sampled trajectories and selection, not a physical duty cycle or volumetric census. RPWS availability varies and CAPS coverage ends in 2012; missing auxiliary data must not become a nondetection. [P7]

Proposed test. Predict the joint magnetic and particle signatures along actual spacecraft trajectories, including quiet eligible intervals and the MAG-based selection rule. Compare occurrence per eligible exposure under different plasma loading and upstream conditions, with epoch-blocked holdouts.

Auxiliary agreement shares the same encounter and is not automatically independent replication. Filamentation mechanisms remain unsettled. In this paper, SCM means Search-Coil Magnetometer; it does not refer to the Spaghetti Cloud Model.

Related established analysis. Xu and colleagues' 1 April 2026 study uses 67 Cassini cusp events and already normalizes occurrence by high-latitude dwell time. Its transport interpretation is informed by MHD simulations with simplified upstream conditions. The useful next step is to reproduce that existing normalization and predict held-out trajectories, not to portray exposure correction as a new omission discovered by this synthesis. [P8]

C6 The Moon provides a calibrated source to record benchmark

Source reported observation. Chang'e-6 NILS measurements from June 2024, published 10 June 2025, significantly detect H⁻. They do not significantly detect O⁻ at the study's 90% threshold. The approximately 10 km layer scale is inferred from photodetachment and transport, not directly mapped. The quoted H⁻ yield of 2.5 percent, with a 90% confidence interval of 1.7 to 3.7 percent, depends on an

angular model and an upstream ARTEMIS proxy. The 302-minute local dataset, slow solar wind and nearby magnetic anomalies limit global extrapolation. [P9]

Source reported inference. Canu-Blot and colleagues' February 2026 author manuscript separates scattering, sputtering and negative-ionization processes by Bayesian inference. It fixes the detector geometric factor; its uncertainty would propagate into inferred ionization. Limited directional coverage, low-energy electron contamination and lander-disturbed regolith also restrict interpretation. The manuscript lists a subsequent A&A publication, whose final publisher version was not independently inspected. [P10]

Exact conditional observation. If the full expected-count model depends on a detector geometric factor and negative-ionization probability only through their product, a compensating change in the two factors preserves its predictions within their physical bounds. Other calibrated channels can break this ambiguity. A measured product is not automatically a calibrated source yield. This is the same elementary identifiability issue encountered in the gamma-ray and grain examples; the actual likelihood and units remain domain specific.

Proposed test. Define the target first: outgoing H⁻ flux, negative-ionization probability, scattering yield or total hydrogen release. Fit energy, direction and time jointly with calibration bounds, electron response, precipitating flux and surface channels. Reserve whole directions or intervals for prediction. The verified public SND dataset contains raw counts summed over surface-pointing directions, so it cannot alone support a directional holdout. Establish the contents and calibration coverage of the fuller ESA archive before promising that reanalysis. [P9, P10]

C7 Lunar source fractions and historical records need distinct denominators

Source reported results. The August 2024 Apollo K/Rb isotope study infers impact vaporization supplies more than 65 percent of atmospheric potassium over long timescales. The February 2026 Kaguya analysis attributes daytime C⁺, N⁺ and O⁺ production mainly to solar-wind sputtering; its extended data use 2008 observations. These claims concern different species, charge states and timescales, so they need not conflict. The latter paper's public abstract, extended-data descriptions and source archive were checked; its main text was access-limited. [P11, P12]

A July 2025 Apollo-material sputtering study provides a further constraint: measured roughness effects and modeled porosity reduce inferred yields relative to common estimates. A modeled highly porous target is not an independent laboratory measurement of that target. Proposed test: compare a universal source mixture with species- and charge-conditioned mixtures, each passed through its own transport and instrument response. Predict held-out day/night or meteor-shower intervals; do not multiply ancient sample and modern spacecraft likelihoods as though they described one contemporaneous event. [P13]

Source reported interpretation. July 2026 Chang'e-6 noble-gas measurements and heating-release patterns are consistent with deeper farside solar-wind implantation than in nearside comparisons. The depth and shielding explanation are inferred, not an ancient particle-speed measurement. A December 2025 Earth-to-Moon transport model also shows that magnetic topology can affect atmospheric transfer and energy selection; its controlled comparison does not support the blanket claim that a magnetic field always increases escape. [P14, P15]

Proposed connection. Treat mineral inventory as a history-dependent record produced by transport, implantation, mixing and extraction. Predict isotope ratios and release steps together while varying exposure age, mineralogy, cosmogenic components and sample handling. Held-out isotopes or heating

steps can test whether a plasma-history explanation outperforms material-history alternatives. Present-day transfer simulations do not uniquely reconstruct a billion-year source history.

C8 Evidence that remains provisional or outside the main claim

The Blue Ghost Lunar Magnetotelluric Sounder provides a promising environment-correction test, but its 2026 conference abstracts give different $+1\sigma$ temperature-contrast bounds at 200 km: below 230 K in LPSC and below 100 K in EGU. Conductivity-to-temperature conversion also assumes a composition, including an Fe profile and treatment of water. These are versioned preliminary inferences, not one settled thermal constraint. Use the PDS instrument metadata for a future data inventory and compare corrected transfer functions across plasma regimes before inferring an interior change. [P16]

The August 2026 lunar radio-occultation manuscript is a useful negative control against geometric overinterpretation. Its line-of-sight spectra cannot uniquely isolate lunar fluctuations from foreground plasma; the reported environment comparison is not statistically robust. A spectral slope is not a direct image of a filament or a measurement of the SCM sideband instability. [P17]

The 25 September 2026 Enceladus-like methanogenesis experiment expands a terrestrial organism's demonstrated laboratory compatibility. Its simulant includes added Fe(II) above predicted natural hydrothermal concentrations. Growth under those controls neither identifies Enceladus methane's origin nor provides a biological prevalence or biosignature-detection likelihood. A future biological source would need the same transport, abiotic-alternative, contamination and response tests as any other proposed source. [P20]

Numerical satellite-collision origin stories, preliminary ring-seismology histories and chemical habitability analogies do not improve the principal SPPT SCM inference here. Their future value depends on an independently discriminating observable. They are not added as evidence for a shared microscopic law, a known historical collision or extraterrestrial life.

C9 Prioritized planetary tests and completion criteria

The Section 7 interpretation and mathematical gates remain in force. The following narrower planetary sequence is ranked by how directly it can remove a present ambiguity. Each starts with a readiness decision, not an assumption that a public archive already supplies the needed record.

- First, specify the lunar negative-ion benchmark. Inventory directional counts, backgrounds, calibration uncertainty and eligible exposure. Proceed to a holdout test only if those records support the proposed target; otherwise finish with the identified product or bound that the available data actually permit.
- Second, complete the Enceladus source and selection readiness check, then compare freezing histories in a parent-to-daughter laboratory test. Require elemental closure, count and mass denominators, paired composition marks and separate actual-impact calibration. Demote unique ocean reconstructions if several histories remain observationally equivalent.
- Third, stress Titan interior families against the complex tidal response and independent rotation or seismic predictions with the appropriate covariance. Reject a family only within the calibrated rheology and structural support actually tested. Reanalysis of the same tracking arcs is not a new independent observation.
- Fourth, turn the ring-age question into a joint source, optical-alteration and material-export test. A clean present inventory alone cannot select an age if unconstrained retention or removal can absorb it. Age inference advances only when those alternatives are independently bounded.
- Use the decagon and magnetospheric surveys as dynamical control cases. Hold out epochs or trajectories and demand predictions of pattern motion or exposure-normalized particle and field

signatures. Do not expand SCM into these domains without a physical state map and a prediction beyond established fluid or plasma baselines.

Final synthesis

This edition closes a bounded research synthesis, not the empirical questions. The defensible advance is a smaller program with explicit targets: preserve the SCM exact graph identities and operational obstructions; give SPPT physically separate inventory, phase and transport accounts; and test each through a calibrated complete observation law. The new planetary evidence makes these requirements more concrete. It does not resolve the SCM quantum completion, validate a new force, determine a unique Saturn ring age or establish a global planetary ontology.

The most informative next success would be one held-out prediction that survives alternative mechanisms and calibrated nuisance support. An honest non-identifiability result is also progress when it specifies the measurement that is missing. Stable SPPT and ASTRA v1.0.7 remains the science authority; this public research draft does not promote or replace it.

Appendix D Planetary source register and access limits

Checked through 5 October 2026. Dates identify publication or stated manuscript versions, not the date every underlying measurement was acquired. Direct locators below support the corresponding P references. An accessible paper, repository landing page or data statement is not a reproduced dataset. No event records or research scripts were executed for this addendum.

[P1] Ricerchi and Crida. Saturn's rings age I Reconsideration of the exposure age. *Icarus* 452, 117029, July 2026. Author manuscript reviewed; published indexed text checked, direct final full-text access incomplete. Optical persistence and retained mass are distinct.
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The Enceladus grain paper, laboratory conference comparison and CDA archive remain E3–E6 in Appendix B. This addendum updates E4 to a freshly read complete conference abstract and E6 to a freshly read complete main article, without upgrading their scientific scope. E3’s final methods, supplement and exact event membership remain unresolved. The source register records what was checked, not blanket verification of every claim made by each cited article.